

4. An owner of a chicken farm would like to test a new feed supplement that is supposed to increase egg production. He randomly selects 8 groups of 100 chickens each and feeds them various levels of the feed supplement. Listed are the amounts of feed supplement and the number of eggs per day.

Group	1	2	3	4	5	6	7	8
Feed supplement	10	10	15	15	20	20	25	25
Number of eggs	78	81	85	79	96	89	93	87

- Construct a scatterplot of the data. Does there appear to be an upward trend that would predict the number of eggs?
- Obtain the least squares estimates of β_0 and β_1 for the model $y = \beta_0 + \beta_1 x + \epsilon$.
- Test the hypothesis $\beta_1 = 0$. Do the results of this test indicate that a linear trend is significant?
- How many additional eggs are expected to be produced for each additional unit of feed supplement?
- Predict the number of eggs if the feed supplement is set at 20 units. Provide the corresponding confidence interval as well.
- Would this model be useful in predicting the number of eggs if the feed supplement is increased to 40 units? Explain.